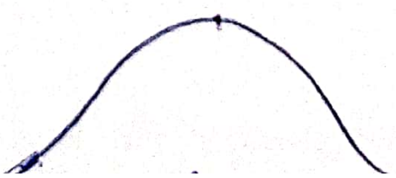
Descriptive Statistics

1st vedio

Measure of Central tendency

Measure of Central tendency are Statistical metrics that describe the Center point or a typical value of a dataset. They Provide a Single Value that Summarizes a Set of data by Identifying the Central position within a dataset.

Ages = [24, 32, 12, 48, 16, 20]



i) Mean
ii) Median
iii) Mode

i) Mean :- Mean is the Sum of all values divided by the No. of Values.

Population mean (μ) | Sample mean ($\bar{x}$)

Population (N) | Sample (n)

$n \leq N$ (: Sin

$N \rightarrow$ Population Size

$$\mu = \sum_{i=1}^{N} \frac{x_i}{N}$$

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

$n \rightarrow$ Sample Size

x is rand. variable

$X = \{5, 8, 12, 15, 20\}$

$N = 5$

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$$\mu = \frac{5+8+12+15+20}{5}$$

$$= \frac{60}{5}$$

$$= 12$$

Chars

→ Affected by Extreme outliers

→ Used for Interval or Ratio Data

$x = \{1, 2, 3, 4, 5\}$

$$\mu = \frac{1+2+3+4+5}{5} = 3$$

$x = \{1, 2, 3, 4, 5, 100\}$

$$\mu = \frac{115}{3} = 23$$

Now use other methods

② Median

The Median is the Middle Value in a dataset when the Values are arranged in ascending or descending order.

$x = \{1, 2, 3, 4, 5\}$

No. of Ele $= 5$

5 is odd

Median $= 3$

$x = \{3, 4, 1, 5, 2, 100\}$

$\Rightarrow \{1, 2, 3, 4, 5, 100\}$

No. $= 6$

6 is Eve

$M = \frac{3+4}{2} = 3.5$

Chars

⇒ Not affected by Extreme outliers

⇒ Used for ordinal, Interval and ratio data.

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③ Mode

Def :- The mode is a value that appears most frequently in a dataset

Data : 2, 4, 4, 6, 7, 7, 7, 9

Mode = 7 (most frequent value)

3, 5, 5, 6, 6, 8

Mode = 5, 6 (Bi modal)

Charteristics

① Not Effected by Extreme values.

② Used for Nominal, Ordinal, Interval and ratio data.

Choosing the Appropriate Measures

Real world Applications

Feature Engineering

Age	weight	Salary	Gender	Degree
24	70	30K	M	BE
25	80	40K	F	-
27	95	50K	F	-
24	-	60K	M	PHD
32	-	-	-	B.E
-	60	-	-	master
-	72	-	-	BSC